of time as a measure of change, as well as the very existence of a kind of cosmic time that allows us to speak sensibly of things like “the age and evolution of the universe as a whole,” rely sensitively on aspects of symmetry. And as scientists have examined that evolution, looking back toward the beginning in search of the true nature of space and time, symmetry has established itself as the most sure-footed of guides, providing insights and answers that would otherwise have been completely out of reach.

Symmetry and the Laws of Physics

Symmetry abounds. Hold a cue ball in your hand and rotate it this way or that—spin it around any axis—and it looks exactly the same. Put a plain, round dinner plate on a placemat and rotate it about its center: it looks completely unchanged. Gently catch a newly formed snowflake and rotate it so that each tip is moved into the position previously held by its neighbor, and you’d be hard pressed to notice that you’d done anything at all. Take the letter “A,” flip it about a vertical axis passing through its apex, and it will provide you with a perfect replica of the original.

As these examples make clear, the symmetries of an object are the manipulations, real or imagined, to which it can be subjected with no effect on its appearance. The more kinds of manipulations an object can sustain with no discernible effect, the more symmetric it is. A perfect sphere is highly symmetric, since any rotation about its center—using an up-down axis, a left-right axis, or any axis in fact—leaves it looking exactly the same. A cube is less symmetric, since only rotations in units of 90 degrees about axes that pass through the center of its faces (and combinations thereof) leave it looking unchanged. Of course, should someone perform any other rotation, such as in Figure 8.1c, you obviously can still recognize the cube, but you also can see clearly that someone has tampered with it. By contrast, symmetries are like the deftest of prowlers; they are manipulations that leave no evidence whatsoever.